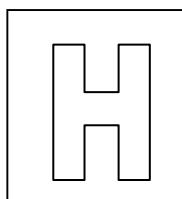


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-University 3

BIOLOGY

9744/02

Paper 2 Structured Questions

14 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the question booklet.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
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7	
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Total	

This question paper consists of 26 printed pages, including 2 blank pages.

[Turn over

Answer **all** questions in this section.

1. The survival and growth of living things are largely affected by the biotic and abiotic factors of the ecosystem. To investigate how temperature of the habitat affects growth of an animal, the relationship between mean mass of different mammalian species and the mean annual temperature of their habitat were investigated. Fig. 1.1 shows the result.

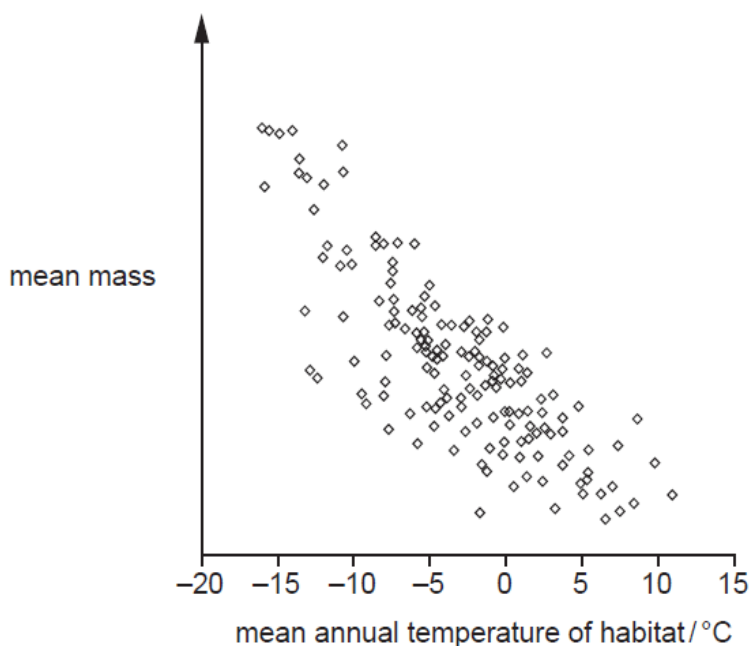


Fig. 1.1

- (a) State what each plot point in the graph represents.

- Individual mammal species;

.....[1]

The mass of a mammal is commonly associated with the mass of fats. The fatty layer under the skin serves many important functions.

- (b) Describe the formation of ester bond in triglyceride.

- Each hydroxyl group (-OH) present in the glycerol molecule reacts with the carboxyl group (-COOH) of a fatty acid molecule;
- For each condensation reaction, a water molecule is removed;

.....
.....
.....

.....[2]

- (c) Discuss the relationship between the mean mass of mammalian species and the mean annual temperature of their habitat.

- The warmer the habitat, the lower mean mass of the mammal species/the colder the habitat, the higher the mean mass of the mammal species;
- Mammals conserve heat using stored fats which takes up mass;
- At each temperature, there is variation/differences in the mean mass among different species;
- Due to different behaviour/activity/diet/shape/amount of stored fat/fur thickness of the mammal;
- The lower the temperature, the greater the need to conserve heat;

.....

[3]

The mean mass of the mammal is studied instead of the individual mass of the mammal, due to the variation in mass between mammals.

- (d) Explain why this is important for survival of species.

- Variation in mass between mammal;
- Natural selection depends on existing variation in a population;
- Genetic variation results in phenotypic variation among organisms within the same population;
- So that under a selection pressure, some phenotype confers selective advantage to certain individuals;
- while others phenotype confers selective disadvantage to other individuals;
- Variations helps to ensure perpetuation/OWTTE of species;
- and safeguard/prevent/OWTTE species from extinction;

2. Since the start of the epidemic, Human immunodeficiency virus (HIV) has caused 35 million deaths worldwide.

(a) Describe the structural components of virus.

- Viral genome contains either DNA or RNA, that can be single or double stranded;
- Genome can be circular or linear, can have more than one copy of the genome;
- All viruses have capsid protein that either surrounds the nucleic acid or encloses it;
- Made from identical protein subunits known as capsomere;
- Capsid protect the viral genome and determine the specific shape of the virus;
- Viral envelopes are phospholipid bilayer membrane derived from the host cell membrane;
- Embedded/OWTTE with glycoprotein and surround the capsid;
- Glycoproteins are essential for the attachment of virus to host cell receptor;
- Protein the virus from enzyme/chemical/harsh environment;

(b) Distinguish between the reproductive cycle of HIV and influenza virus.

Feature	HIV	Influenza
Entry	HIV enters host cell via <u>fusion</u> between membrane of influenza and cell surface membrane of host cell'	Influenza enters host cell via <u>endocytosis</u>
Target cell	CD4+ T helper cell/macrophage	<u>Epithelial cell</u> of respiratory tract
Genome	8 single stranded negative sense RNA	2 single-stranded positive sense RNA
Reverse transcriptase	Reverse transcription of RNA to dsDNA	No reverse transcription
Integration of viral genome	Integration of HIV dsDNA into host chromosome	No integration of viral genome
Proteolysis of viral protein	Proteolysis of viral polyprotein to form mature viral proteins;	No proteolysis;
Latency phase	Latency phase in HIV life cycle;	No latency period in influenza;

.....[3]

(c) Explain how HIV infection causes disease in humans.

- HIV causes Acquired Immunodeficiency Syndrome (AIDS);
- Target cells include CD4+ T helper cells and macrophages;
- The window period/primary infection is the incubation period of HIV, where the infected individual display/exhibit no symptoms/physiological response;
- Followed by the **acute primary infection phase** where the infected individual display symptoms;
- Decrease number of CD4+ T helper cells and increase in free HIV particles;
- HIV particles decrease is followed by the **asymptomatic phase** where;
- Gradual drop/decline in the number of CD4+ T helper cell and symptoms disappear;
- Immune system can still manage the free viruses and infected cells;
- **Symptomatic phase** is characterised by immune suppression and onset of symptoms;
- Decrease in number of CD4+ T helper cells and increase in free HIV particles;
- Leading to AIDS where the immune system fails and opportunistic infections and cancer occur;

[4]

One common treatment method for HIV infected patients is the use of reverse transcriptase inhibitors. These inhibitors bind to the inhibitor binding site on the reverse transcriptase.

However, HIV often develops resistance against these inhibitors due to a single base substitution in the DNA sequence, resulting in a change of an amino acid in a subunit of reverse transcriptase. The mutant protein has the same number of amino acid sequence as the wild type protein and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

(d) Explain why the number of amino acid in mutated protein remains unchanged and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

Same number of amino acid

- DNA mutation result in a missense mutation, and not frame-shift mutation;

Retains reverse transcriptase function

- Active site of enzyme is not altered/OWTTE after mutation;

Does not allow binding of inhibitor

- Inhibitor binding site is no longer complementary to the 3D conformation/shape of the inhibitor;
- Change in amino acid sequence results in change in tertiary structure/3D conformation of protein;
- AVP;

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..[3]

[Total:13]

3. Fig. 3.1 shows a plant cell undergoing gametogenesis.

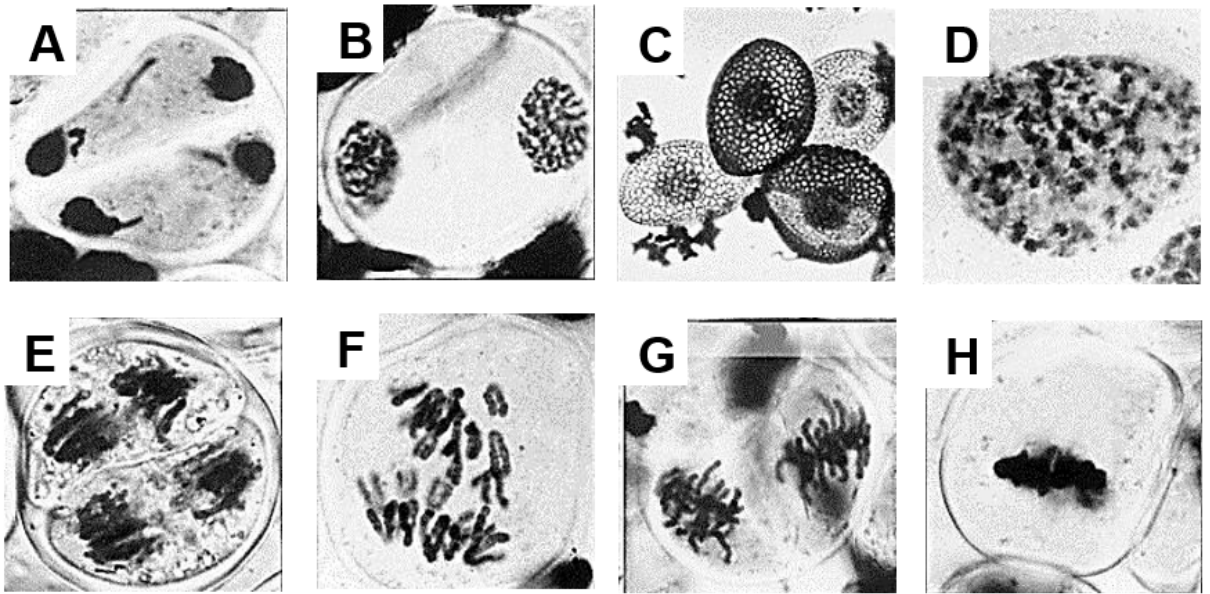


Fig. 3.1

- (a) Arrange the letters **A** to **H** to show the correct sequence of events for this nuclear division.

• D H F **B** G E A C;

.....[2]

The cell was exposed to a toxin that causes spindle fibres to increase in length.

- (b) Explain how the cell will be affected if the toxin was administered towards the end of stage H.

- The interaction between centromere and spindle fibres will be disrupted;
- Metaphase plate may be disorganised/OWTTE;
- Anaphase I may not occur and homologous chromosome fail to separate;
- Length of cell may increase as pole to pole spindle fibre increases;
- AVP

[3]

- (c) Explain how the toxin may lead to an absence of daughter cells or daughter cells with varied genetic content.

- M phase checkpoint may be triggered;
- to cause cell cycle arrest resulting in absence of daughter cells;
- failure of checkpoint resulting in daughter cells with unequal number of chromosome; due to non-disjunction;
- AVP;

[2]

- (d) Explain the significance of this nuclear division during gametogenesis.

- Meiosis is a reduction division which is essential in producing haploid gametes that have half the number of chromosomes as an original somatic cell;
- In order to restore the original diploid / chromosome number in a zygote when gametes fuse;
- Meiosis creates genetic variation, essential for evolution;
- Ensures that species constantly change and adapt when environmental conditions change;
- AVP;

[3]

[Total:10]

4. During an investigation, pollen grains from the anthers of a pure-bred plant with green leaves and hairy fruit were transferred by a student to the stigmas of a pure-bred plant with mottled (green and yellow) leaves and smooth-surfaced fruits. All the F₁ generations had green leaves and smooth fruit.

(a) Explain what is meant by pure-bred

- Refers to alleles which are homozygous;

.....
...[1]

(b) Explain how the genotype is linked to phenotype in tomato.

- Genotype is the genetic make-up of an organism;
- Expression of the gene leads to the observable or measurable trait;
- Phenotype is influenced by both genetic and environmental factors;

.....
...[2]

A test cross was then made between the F₁ plant and a plant with mottled leaves and hairy fruit. The phenotype of 100 offsprings of the test cross were recorded and shown in Table 4.1.

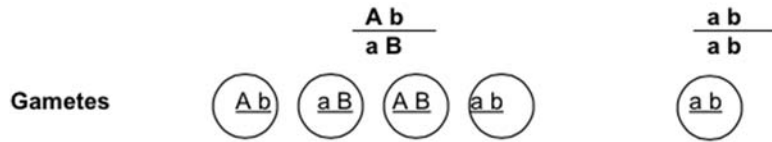
Table 4.1

Green leaves, smooth fruit	Green leaves, hairy fruit	Mottled leaves, smooth fruit	Mottled leaves, hairy fruit
8	46	40	6

- (c) Using suitable symbols, draw a genetic diagram to explain the results of the test cross.

Let **A** represent the **dominant allele** for **green leaves**, **a** represent the **recessive allele** for **mottled leaves**.
B represent the **dominant allele** for **smooth fruit**, **b** represent the **recessive allele** for **hairy fruit**.

F₁ test cross Green leaves, smooth fruit x Mottled leaves, hairy fruit



Random Fertilization

		Female Gametes			
		(A b)	(a B)	(A B)	(a b)
Male Gametes	(a b)	$\frac{A\ b}{a\ b}$	$\frac{a\ B}{a\ b}$	$\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$
		Green leaves, hairy fruit 46	Mottled leaves, smooth fruit 40	Green leaves, smooth fruit 8	Mottled leaves, hairy fruit 6
		Parental		Recombinants	

- F1 genotype;
- Gametes;
- Results of punnett square;
- F2 genotype;
- F2 phenotype and phenotypic ratio;

(d) Explain the difference between the observed test cross results from the expected test cross ratio.

- Expected test cross ratio is 1:1:1:1 while observed test cross results is not;
- Indicates that the 2 genes for leaves colour and fruit texture are linked/on the same chromosome;
- Therefore, there is crossing over between the 2 linked genes in some of the cells;
- And thus the observed offspring of test cross are predominantly/majority of parental combination/ORAs;

.....
.....
.....
..[2]

In a separate investigation, a test cross was performed between the same F1 plant and plant with mottled leaves and hairy fruit. Instead of recording the results for all four groups of phenotypes, the student recorded the number of the two more common groups of phenotype among the 100 offsprings. The results were shown in Table 4.2.

Table 4.2

Green leaves, smooth fruit	Green leaves, hairy fruit	Mottled leaves, smooth fruit	Mottled leaves, hairy fruit
<i>Not recorded</i>	52	44	<i>Not recorded</i>

The student claimed that the observed test cross ratio of offspring can be simplified to 1:1.

(e) Suggest why the student made this claim.

- The 2 genes for leaf colour and fruit texture are very closely linked;
- Crossing over between the 2 genes is very rare/low probability;
- The 2 genes are inherited as an unit/do not show independent assortment during meiosis;

.....
[1]

- (f) Perform the chi-squared test to verify this claim.

The formula and probability table is given as follow.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degrees of freedom	Probability		
	0.10	0.05	0.01
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

- $[(52-48)^2 + (44-48)^2]/48 = 0.67$;
- At $p = 0.05$, degree of freedom = 1, critical χ^2 value is 3.84, since calculated chi-square value is less than critical chi-square value;
- There is no significant difference between the test cross and expected 1:1 ratio and hence student's claim is valid;

.....

[3]

[Total:14]

5. Fig. 5.1 shows an electron micrograph of an organelle.

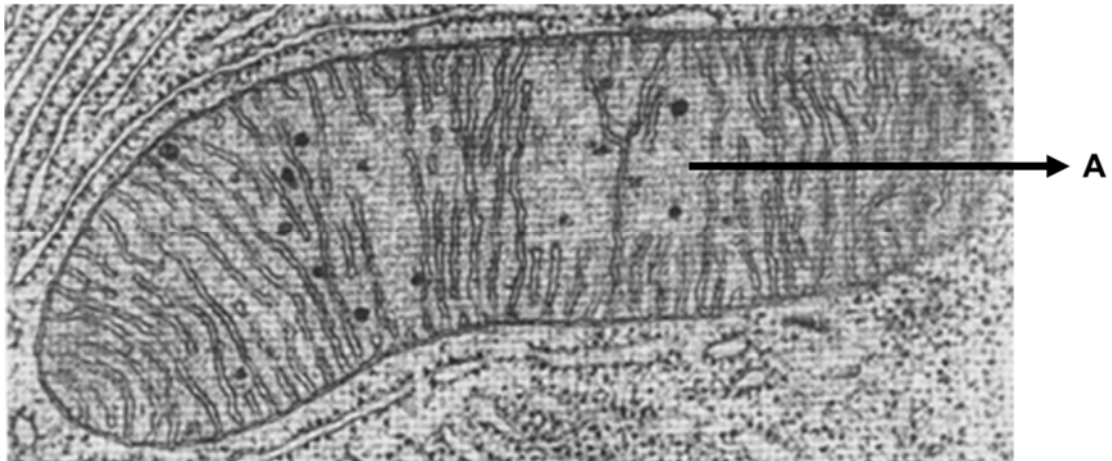


Fig. 5.1

- (a)** State the stage(s) of aerobic respiration that occur in **A**.

- Link reaction and Krebs cycle;

....[1]

- (b)** List down the end products of the stage(s) identified in **(a)**.

- Link reaction: acetyl coa, NADH and CO₂;
- Krebs cycle: regeneration of oxaloacetate, NADH, FADH₂, ATP and CO₂;

.....
....[2]

- (c)** Describe the importance of membrane during aerobic respiration.

- The outer mitochondrial membrane compartmentalises/separate content of organelle from cytoplasm for aerobic respiration to occur;
- The inner mitochondrial membrane/cristae contains electron transport chain and ATP synthase to carry out oxidative phosphorylation;
- The inner mitochondrial membrane/cristae is highly folded and this increases the surface area to hold the enzymes;
- The inner mitochondrial membrane is impermeable to protons and thus act as barriers/OWTTE to the movement of protons;
- Maintains concentration gradient of protons between intermembrane space and matrix;
- Allowing generation of the proton motive force across the cristae for chemiosmosis to occur;

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....[3]

Anaerobic respiration occurs in animals, plants, bacteria and archaea. It occurs when there is an absence of oxygen, releasing less amount of energy as compared to aerobic respiration.

(d) Explain how anaerobic respiration releases energy.

.....

- | | |
|---|--|
| <ul style="list-style-type: none"> • <u>Regenerating NAD⁺ for glycolysis;</u> • <u>Glycolysis synthesise two net ATP molecule per glucose breakdown;</u>
[Compulsory] • <u>Lactate fermentation is the reduction;</u> • <u>of pyruvate to lactate catalysed by lactate dehydrogenase.;</u> • <u>Alcoholic fermentation is the decarboxylation;</u> • <u>of pyruvate to ethanal catalysed by pyruvate decarboxylase;</u> • <u>Ethanal is then reduced to ethanol, catalysed by alcohol dehydrogenase;</u> | <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> |
|---|--|

.....[4]

- (e) Many enzymes are involved in the different stages of aerobic and anaerobic respiration.

Fig. 5.2 shows the changes in energy during the progress of an uncatalyzed reaction.

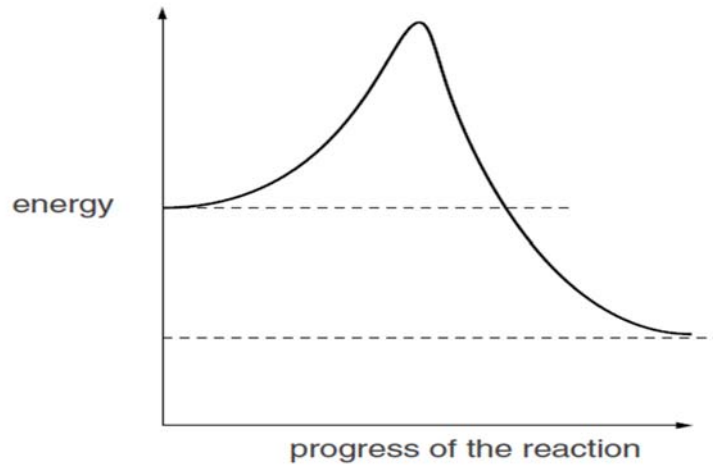


Fig. 5.2

- (i) Draw on Fig. 5.2 the changes in energy during enzyme-catalysed reaction.

- Correct starting and ending level;
- Lower activation energy;

[1]

- (ii) Label the following terms on your drawn diagram.

- Correct drawn label for activation energy;
- Correct drawn label for overall energy released;

[1]

[Total:12]

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6. Fig. 6.1 shows the effects of temperature and light intensity on the rate of photosynthesis of a plant.

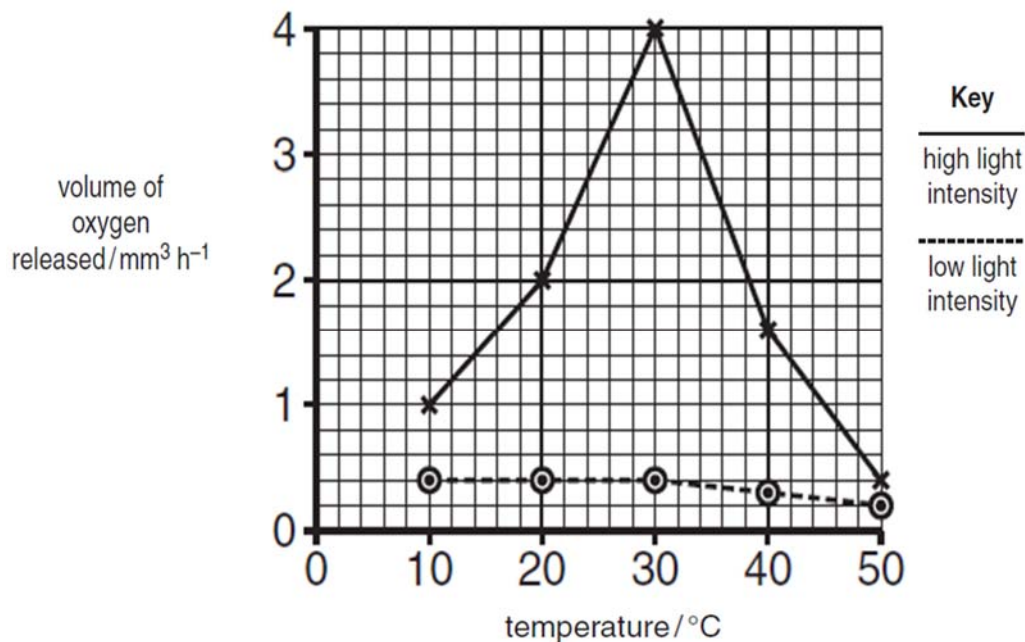


Fig. 6.1

- (a) Compare the effect of temperature on the rate of photosynthesis between low and high light intensity.

- As temperature increases from 10°C to 30 °C, the volume of oxygen released increases sharply/steeply/significantly from 1 mm³h⁻¹ to 4 mm³h⁻¹ at high light intensity but the volume of oxygen released remains constant at 0.2 mm³h⁻¹ at low light intensity;
- As temperature increases from 30 °C to 50 °C, the volume of oxygen released decreases sharply/steeply/significantly from 4 mm³h⁻¹ to 0.2 mm³h⁻¹ at high light intensity while the volume of oxygen released decreases slightly from 0.2 to 0.1 mm³h⁻¹ at low light intensity;
- The volume of oxygen released/rate of photosynthesis reaches the maximum/peak at 30 °C for high light intensity but the volume of oxygen released/rate of photosynthesis did not reach any maximum/peak at low light intensity;

[3]

(b) Account for your answer in (a).

- | | |
|--|------|
| • At high light intensity, as temperature increase from 10°C to 30 °C, the <u>kinetic energy</u> of molecules <u>increases</u> ; | |
| • Chances of <u>effective collision</u> between molecules is at <u>higher frequency</u> , leading to <u>greater/higher rate of photosynthesis</u> ; | |
| • At higher rate of photosynthesis, <u>volume of oxygen released increases</u> due to <u>photolysis of water</u> ; | |
| • At low light intensity, as temperature increase from 10°C to 30 °C, the rate of photosynthesis is <u>limit by the amount of light intensity/limiting factor is light intensity</u> ; | |
| • At low and high light intensities, as temperature increase from 30°C to 50 °C, the <u>proteins</u> are <u>denatured</u> and became non-functional; | |
| • <u>Lesser enzyme-substrate complexes formed per unit time</u> ; | |
| • Lesser <u>photolysis</u> occurs and thus <u>lesser oxygen</u> released; | |

[5]

[Total:8]

7. Recent research has revealed connection between the size of the nucleolus and the lifespan of the cell. Researchers believed that the nucleolus is an important organelle that coordinates many processes. This may imply that it is involved in many different signalling pathways.

(a) Suggest why the size of the nucleolus varies in different tissues.

- The nucleolus is the site for transcription of rRNA;
 - and the assembly of ribosomal subunits;
- Max 1**
- Tissue such as muscle/skin/AVP with higher metabolic activity has bigger nucleolus/tissue with low metabolic activity such as neurones with smaller nucleolus;

[2]

Reduced insulin signalling, dietary restriction and other signalling pathways increase the activity of a gene known as *NCL-1*, a nucleolar regulating gene. The increase in gene expression of *NCL-1* is associated with reduced size of nucleolus and increased lifespan.

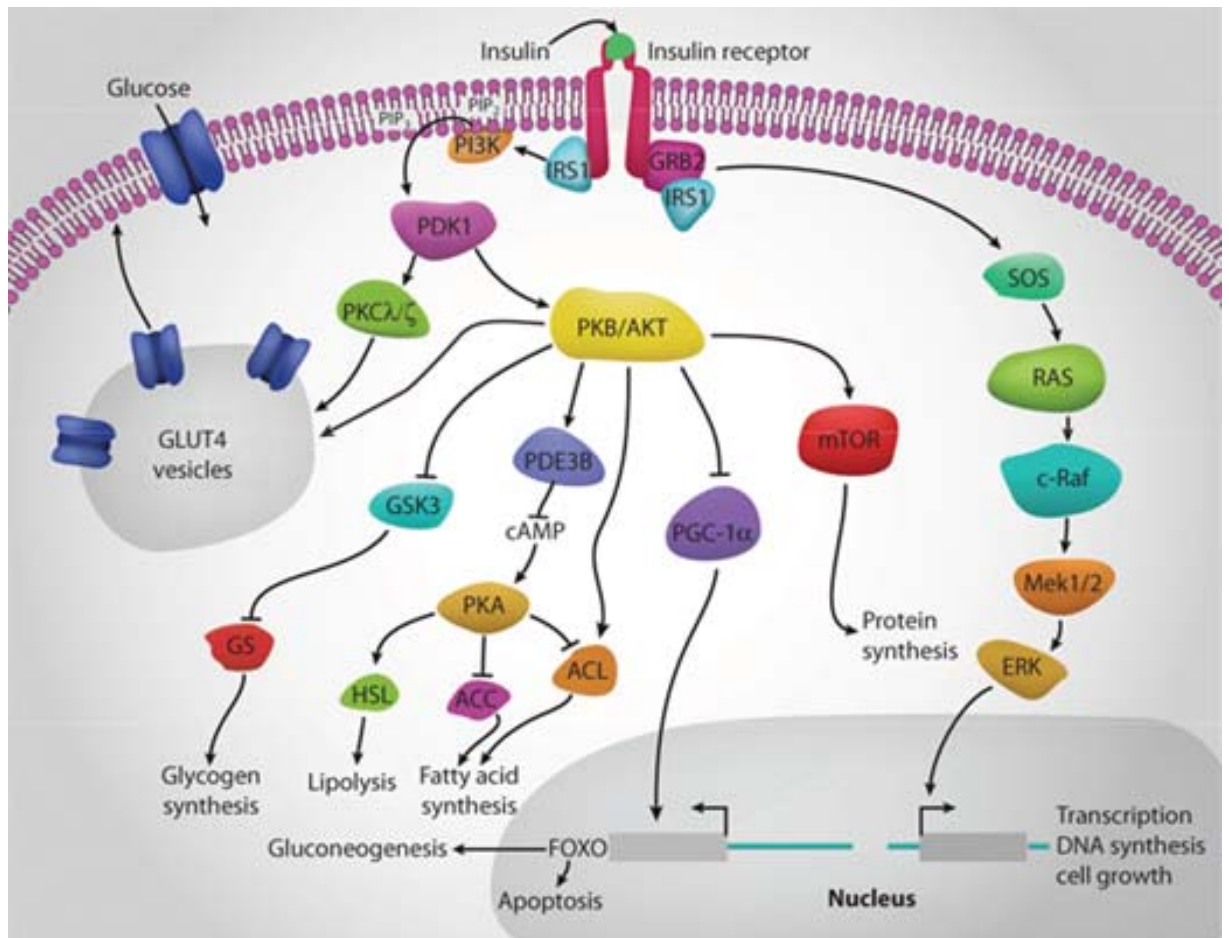
(b) Describe how expression of *NCL-1* gene can be increased at the chromatin level in eukaryotes.

- Chromatin remodelling complex alter the structure of nucleosome to be less tightly bound/OWTTE to histone;
- Allow RNA polymerase and transcription factor to access promoter easily;
- Demethylation/removal of methyl group from DNA;
(Allow RNA polymerase and transcription factor to access promoter easily)
- Histone acetylation will loosen/OWTTE the ionic bond between DNA and histone;
- Addition of acetyl group to positively charged lysine in histone tails;
- Catalysed by histone acetyl transferase;
- Acetylated lysine acts as binding sites for chromatin remodelling complexes;
- Demethylation/removal of methyl group from histone;

[4]

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Insulin has several effects on cellular growth and metabolism. The actions of insulin are mediated by the insulin receptor, and the end results vary in different cells. Fig. 7.1 shows the signalling pathways activated when insulin binds to the receptor.



Michael W.King: Integrative Medical Biochemistry

Fig. 7.1

The association between insulin signalling and cancer risk is widely debated by scientists due to the release of contradicting studies.

(c) With reference to Fig. 7.1, suggest how dysregulation of the insulin signalling pathway may lead to cancer.

- | | |
|---|----------|
| <ul style="list-style-type: none"> Increased synthesis/activation of SOS/RAS/c-RAF/Mek1/2/ERK leads to increased expression of gene involved in DNA synthesis and cell growth; | |
| <ul style="list-style-type: none"> Increased synthesis/activation of PGC/FOXO leads to increase rate of gluconeogenesis, providing the cell with more glucose to cope with the increase metabolism rate during cancer; | |
| <ul style="list-style-type: none"> Decreased synthesis/activation of PKB/AKT leads to increase rate of gluconeogenesis, providing the cell with more glucose to cope with the increase metabolism rate during cancer; |[3] |
| <ul style="list-style-type: none"> Decrease synthesis/activation of PGC/FOXO leads to decrease rate of apoptosis, inhibiting the death of cancerous cell; | |
| <ul style="list-style-type: none"> Increase synthesis/activation of PKB/AKT leads to decrease rate of apoptosis, inhibiting the death of cancerous cell; | |
| <ul style="list-style-type: none"> Mutation of RAS; | |
| <ul style="list-style-type: none"> AVP; | |

(d) Suggest why dysfunctional insulin receptors is linked to many abnormalities.

- | | |
|--|------|
| <ul style="list-style-type: none"> Activated <u>second messengers</u> recruit and activate many other downstream relay protein and <u>protein kinase</u>; | |
| <ul style="list-style-type: none"> Each activated protein kinase will initiate a sequential phosphorylation and activation of other kinases/<u>phosphorylation cascade</u>; | |
| <ul style="list-style-type: none"> At each phosphorylation step, <u>each activated kinase</u> is able to activate a <u>large number of the next kinase</u>; | |
| <ul style="list-style-type: none"> <u>Signal amplification</u> result in activation of <u>large</u> number of different effector protein; | [3] |
| <ul style="list-style-type: none"> Large number of effector proteins serving different cellular functions; | |

12]

8. Tuberculosis is an infectious disease caused by the intracellular pathogen, *Mycobacterium tuberculosis*. The disease affects the respiratory tract and caused an estimated 1.6 million deaths in 2016. Tuberculosis is treated with antibiotics such as rifampicin.

Increasingly, bacteria are developing resistance towards rifampicin. This is because bacteria contain an enzyme, rifampicin monooxygenase, which changes the structure of rifampicin from cyclic to linear, and thus inactivating the antibiotics.

(a) Define antibiotics.

- | | |
|---|--------|
| • Antibiotics are <u>natural substances</u> derived from <u>microorganism/bacteria/fungi</u> ; | |
| • that <u>inhibit or kill</u> bacteria by <u>disrupting/OWTTE the structure</u> and <u>metabolism</u> of <u>bacterial</u> cell; | |
| • <u>Bacteriostatic</u> antibiotics <u>inhibit the growth and cell division</u> of bacteria; | |
| • <u>Bactericidal</u> antibiotics <u>kill</u> bacteria when the bacteria are in the process of undergoing cell division; | ...[2] |

(b) Suggest how scientists should design a new drug to overcome the problem with rifampicin resistance.

- | | |
|--|----|
| • Design new rifampicin analogs that are not susceptible to the enzyme inactivation; | .. |
| • Design drugs that inhibit the enzyme; | |
| • AVP; | |
- [1]

LRRK2 is a large protein encoding seven functional domains as shown in Fig. 8.1. LRRK2 has 2527 amino acids and a molecular weight of 286 kDa. It is a multi-domain protein comprising of a catalytic core domain, kinase domain and five protein-protein interaction domains. Mutations in the gene encoding for LRRK2 are associated with Parkinson's disease, cancer, Crohn's disease and bacterial infections.

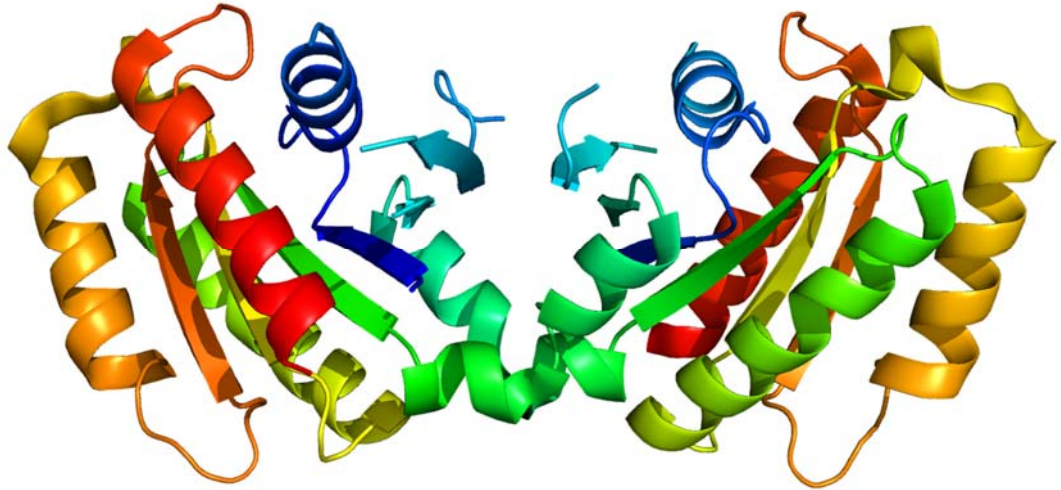


Fig. 8.1

(c) Using the information above, explain why there are more than one disease associated with mutations in the gene coding for LRRK2.

- Mutation result in the change in sequence of base of any of the domain may lead to change in sequence of mRNA;
 - Leading to a change in sequence of amino acid in polypeptide;
 - Eventually leading to change in property and function of the protein;
- Max 2**
- Mutation in the protein-protein interaction domains may alter the interaction between LRRK2 and target protein;
 - affecting the 3-dimensional conformation and function of the target protein;
 - Mutations in the catalytic core domain may alter the catalysis of target substrate;
 - Mutations in the kinase domains may alter the phosphorylation of target protein;

[4]

Studies have shown that LRRK2 inhibits the maturation of phagosome in macrophages.

(d) Suggest what will happen when an individual infected with *Mycobacterium tuberculosis* have functional LRRK2 protein.

- As *M. tuberculosis* enters the lungs, alveolar macrophages phagocytose the mycobacteria; ..
- And attempts to form a phagosome containing the bacteria/ *M. tuberculosis*; ..
- The mycobacterium will fail/OWTTE to inhibit the fusion of phagosome with lysosome; ..
- Phagolysosome is formed and lysosomal enzyme will kill the bacteria/ *M. tuberculosis*; ..
- *M. tuberculosis* fails to survive /die and infection will stop/OWTTE; ..

.....[3]

Fig. 8.2 shows the organisation of the DNA sequence which encodes one of the domains of human LRRK2. The size of the exon and intron is stated within the brackets.

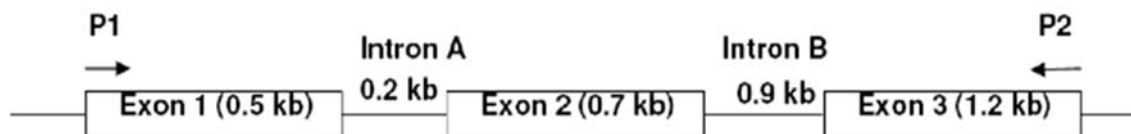


Fig. 8.2

The DNA sequences from three individuals, normal, mutant **A** and mutant **B** were obtained. The DNA sequences were subjected to transcription to form mRNA. The mRNA then underwent reverse transcription to form complementary DNA (cDNA). Polymerase chain reaction (PCR) was performed using primer **P1** and **P2**, which were specific to the exon regions shown in Fig.8.2.

(e) Explain why cDNA sequences were used instead of DNA sequence.

- cDNA sequences are complementary to the mRNA sequence/exact copies of DNA sequence without the introns;
- the mRNA went through pre-mRNA splicing to remove/splice the introns;

.....
.....

.....[2]

(f) Explain why two primers are needed for the polymerase chain reaction.

- Serve as the forward and reverse primers;
- Each primer is complementary and binds to the 3' end of each template strand of the double helix;

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.....[2]

The products of PCR were subjected to electrophoresis on an agarose gel and the results are shown in Fig. 8.3.

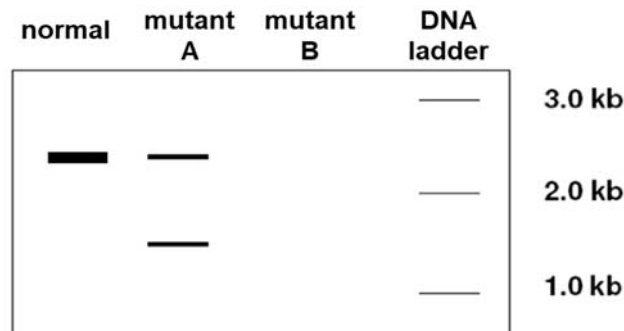


Fig. 8.3

- (g) Deduce the size of the PCR product for the normal individual. Show your working in the space below.

$$0.5 + 0.7 + 1.2 = 2.4\text{kb};$$

[1]

- (h) Account for the band patterns observed for

(i) mutant A

- mutant A has one band pattern same as normal individual/2.4km and another band pattern of 1.7kb ;
- implying that this person is heterozygote with one normal allele and one mutant allele;
- mutant allele is a result of deletion of exon 2, intron A and intron B;

.....

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....[2]

(ii) mutant B

- No bands because there is no PCR product;
- Due to mutation in the binding site of primer P1 and/or P2;
- When primer P1 and/or P2 cannot bind, initiation of DNA replication cannot occur;
- OR
- No bands because transcription/reverse transcription did not occur;
- Due to mutation in the promoter/ general transcription factor/reverse transcription start site;
- AVP;

.....

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.....

...[2]

(i) Explain why there are more than one bands in the DNA ladder.

- DNA ladder is a mixture of DNA fragments of various sizes;
- Used to estimate the size of DNA fragments in sample;
- Gel electrophoresis separates DNA fragment by size/molecular weights where smaller fragments move faster and further from the starting point/ bigger fragments move slower and closer to the starting point;

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.....[2]

[Total:21]

End of Paper